FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS
FULL TEST– V
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	B	B
2.	A	C	A
3.	D	B	C
4.	B	A	D
5.	A	B	C
6.	B	A	C
7.	A	C	B
8.	C	A	D
9.	A, C	B, C	A, C
10.	A, D	B, C	A, D
11.	B, C	A, B, C, D	A, C
12.	A, B, D	A, B, C	C, D
13.	C	D	B
14.	C	A	A
15.	B	C	D
16.	D	A	D
17.	A	A	C
18.	C	B	B
1.	3	2	8
2.	2	4	4
3.	6	0	4
4.	6	1	6
5.	2	4	1

Physics

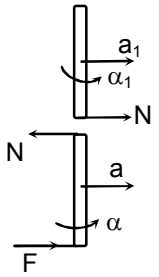
PART – I

SECTION – A

1. $V_{CM} = \dfrac{m_1v_1 + m_2v_2}{m_1 + m_2} = 1\,m/s$

At maximum extension both blocks will move in same direction with V_{CM} now use energy y conservation.

2. $(F + N)\dfrac{\ell}{2} = \dfrac{M\ell^2}{12}\alpha$



$F - N = Ma$

$N = Ma_1$

$\Rightarrow N\dfrac{\ell}{2} = \dfrac{M\ell^2}{12}\alpha_1$

Acceleration of hinge O

$= a - \dfrac{\ell}{2}\alpha = a_1 + \dfrac{\ell}{2}\alpha_1$

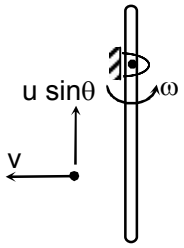
$\Rightarrow N_x = -F/4 \Rightarrow a = 5F/4M$ and $\alpha = 9F/2M\ell$

$\Rightarrow \vec{a}_o = \left(a - \dfrac{\ell}{2}\alpha\right)\hat{i} = -\dfrac{F}{M}\hat{i}$

3. $y = y_0 \sin kx \cos\omega t$ and $\dfrac{9\lambda}{4} = L$

4. $\dfrac{d\theta}{dt} = \dfrac{KA}{L}(T_1 - T_2) \rightarrow$ equation of heat flow through the box.

5. Just after collision.

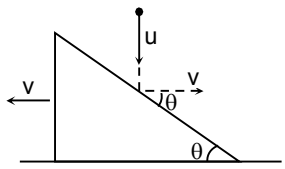


$\dfrac{\omega(a+b) + v}{u\cos\theta} = e$

$m_2(v + u\cos\theta)(a+b) = m_1\left(\dfrac{L^2}{12} + a^2\right)\omega$

6. $\dfrac{1}{2}LI^2 + \dfrac{\theta^2}{2C} = \text{constant}$

7. Using COM;



$v\cos\theta = u\sin\theta$

and $e = \dfrac{(2v\sin\theta)}{u\cos\theta} = 2\tan^2\theta$

$\Rightarrow 2\tan^2\theta < 1$

$\Rightarrow \tan\theta < 1/\sqrt{2} \Rightarrow \theta < \tan^{-1}\left(\dfrac{1}{\sqrt{2}}\right)$

8. Power factor = $V_{RMS}\, I_{RMS} \cos\phi$

9. For maxima $y = \frac{n\lambda p}{d}$ for minima $y = (2n-1)\frac{\lambda p}{2d}$

10. $y = A\sin(\omega t - kx)\frac{\delta y}{\delta t} = A\omega\cos(\omega t - kx)$

11. $V_{medium} = \frac{C}{\mu}$

12. Isothermal $PV = constant$ adiabatic $PV^{r} = constant$

13-14. $h\nu = \phi + eV_0$

15-16. $E = \Delta mc^2$

17-18. $f = \frac{C \pm V_0}{C \mp V_S} \times f_0$

SECTION – C

1. $3\sqrt{2}$ km/hr

2. The centre of mass of system will not move horizontally.

3. $F = U_{Rel}\frac{dM}{dt}$

4. In steady state the current in branch containing capacitor will become zero.

5. $F = q\left[\vec{E} + \vec{V} \times \vec{B}\right]$

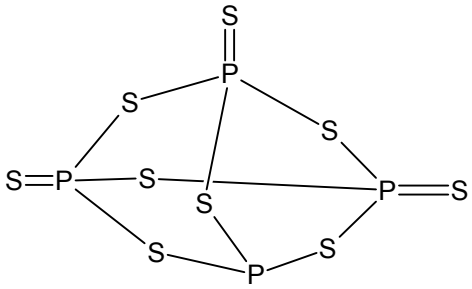
Chemistry

PART – II

SECTION – A

1. $\left[\overset{-1}{N}H_2OH - H\right]^+ \longrightarrow \overset{+1}{N}_2O + 2\overset{-3}{N}H_4 + 2H^+ + 3H_3O^+$

2.



Has 3 $p\pi$ - $d\pi$ P – S bond
Has 3 P = S bonds
Has 3 tetrahedral units of P
Has 6 P – S – P linkages.

3. $KI + 2KMnO_4 + H_2O \longrightarrow KIO_3 + 2KOH + MnO_2$

(n_f = 6) (n_f = 3)
Moles of KI × 6 = 3 × 0.2 × 10 × 10^{-3}
Moles of KI = 10^{-3} = 1 milli mole = x

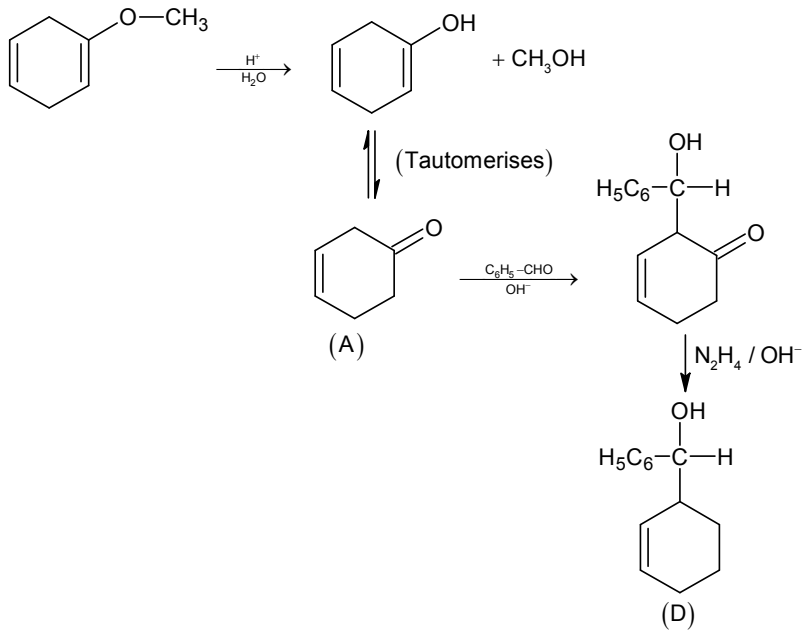
$\bar{S} - C \equiv N + I_2 + H^+ \longrightarrow SO_4^{2-} + HCN + I^- + H_2O$

To produce 10^{-3} moles I^-,

Moles of $\bar{S}CN \times 6 = 10^{-3} \times 1$

Moles $\bar{S}CN = \dfrac{10^{-3}}{6} = 1.6 \times 10^{-4}$ moles

4.



2 enantiomeric pairs possible for (D).

5. $CoCl_2 + 2KCN \longrightarrow Co(CN)_2 \downarrow + 2KCl$

Buff coloured

$Co(CN)_2 + 4KCN \longrightarrow K_4[Co(CN)_6]$

6.

	Half cell	Reaction E^o
L.H.S.	$H_2 \longrightarrow 2H^+ + 2e^-$	$E_{ox^o} = 0.00$ V
R.H.S.	$Zn^{2+} + 2e^- \longrightarrow Zn$	$E^o_{red} = 0.76$ V

Net $\Rightarrow$ $zn^{2+} + H_2 \longrightarrow Zn + 2H^+, E^o_{cell} = 0.76$ V

$$K = \frac{[H^+]^2}{[Zn^{2+}]} = \frac{0.0591}{n}\log K$$

$$E_{cell} = E^o_{cell} - \frac{0.0591}{n}\log K$$

$$-0.46 = -0.76 - \frac{0.0591}{2}\frac{\log_{10}[H^+]^2}{0.3}$$

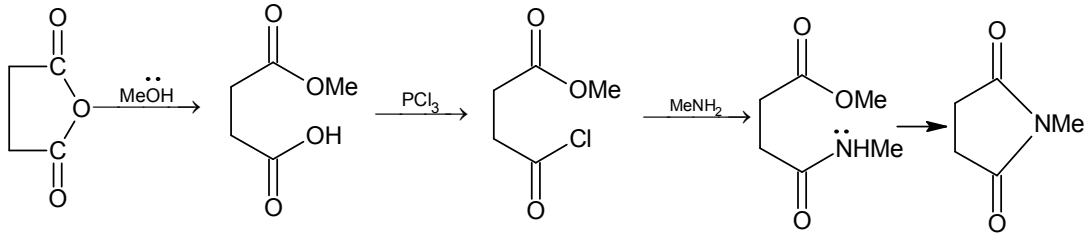
$$[H^+] = 4.60 \times 10^{-6}\ M$$

$\Rightarrow$ $HSO_3^- \rightleftharpoons H^+ + SO_3^{2-}$

$0.4\ M \quad 4.6 \times 10^{-6} \quad 6.4 \times 10^{-3}$

$$K_a = \frac{[H^+][SO_3^{2-}]}{HSO_3^-} = \frac{4.6 \times 10^{-6} \times 6.4 \times 10^{-3}}{0.4} = 7.3 \times 10^{-8}$$

$pK_a = 7.13$

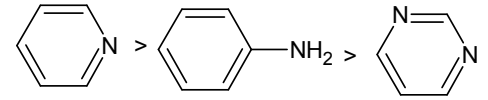
7. 

8. Shortest λ of Lyman series $(\infty \to 1)$ for H-atom $\frac{1}{\lambda_1} = R_H\left(\frac{1}{1} - \frac{1}{\infty^2}\right) \Rightarrow \lambda_1 = R_H$

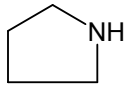
9. (2) $B(OH)_3 + NaOH \rightleftharpoons NaBO_2 + Na[B(OH)_4] + H_2O$ can be made to proceed forward by adding *cis-diol* to stabilization by chelation.
(3) The solution turning milky on passing H_2S through group II solution indicates the presence of oxidizing agent where H_2S is oxidised to $8O_2$.

10. (A) The heat of formation for NO_2 is calculated as – 45 kJ mol^{-1}.
(B) For reaction $N_2 + 2O_2 \longrightarrow 2NO_2$, $\Delta H = -310$ kJ / mole. i.e. it is exothermic.
(C) $\frac{1}{2}N_2 + \frac{1}{2}O_2 \longrightarrow NO,\ \Delta H^o = -55$ kJ / mol
(D) $NO + \frac{1}{2}O_2 \longrightarrow NO_2,\ \Delta H^o = -100$ kJ / mol, it is exothermic.

11. (A) No H-bonding can occur in chlorobenzene, so +I effect increases acidic strength.
(B) Due to both steric and polar effect has no ortho effect is observed in case of o-$MeOC_6H_4NH_2$.
$MeOC_6H_5NH_2$.
p-$MeOC_6H_4NH_2$ > o-$MeOC_6H_4NH_2$ > m-$MeOC_6H_4NH_2$
$pK_a = 5.24$ ($pK_a = 4.45$) ($pK_a = 4.2$)

(C)  > >

$pK_a = 5.25$ $pK_a = 4.63$ $pK_a = 1.3$

(D)  > $(CH_3CH_2)_2NH$

Pyrrolidine
(pKa = 11.27) (pKa = 10.98)

12. (A) is correct, $\Delta S_{sys} = nR\ell n\frac{V_2}{V_1}$, $V_2 > V_1$.

$\therefore \Delta S_{sys} = +ve$

(B) PV^{γ} = constant, as γ is high, slope is higher.
(D) At boiling point, the process is at equilibrium $\therefore \Delta G = 0$.

14. x = 6, y = 3, z = 2
Solution for the Q. No. 13 & 14.

$$IO_3^- + 6OH^- + Cl_2 \longrightarrow IO_6^{5-} + 3H_2O + 2Cl^-$$

$$\Rightarrow IO_6^{5-} \xrightarrow{H^+} H_5IO_6$$

$$\Rightarrow 2H_5IO_6 \xrightarrow[-4H_2O]{100^\circ C} 2HIO_4 \xrightarrow{200^\circ C} I_2O_5.$$

17. V. D. = 28.75

$$\alpha = \frac{D-d}{(n-1)d} = 0.6$$

$$N_2O_4 \rightleftharpoons 2NO_2$$

At equilibrium: $(1-0.6)$ 1.2

$= 0.4$

$$\frac{P_{N_2O_4}}{P_{NO_2}} = \frac{0.4}{1.2}$$

If moles of N_2O_4 escaping out is x, then for NO_2 it is (1 – x)

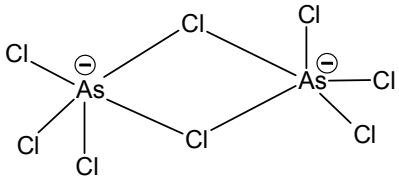
$$\left(\frac{x}{1-x}\right) = \frac{0.4}{1.2}\sqrt{\frac{46}{92}} \Rightarrow x = 0.188$$

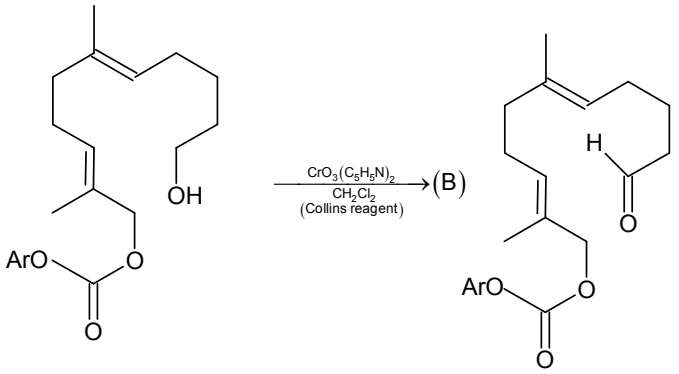
$X_{N_2O_4(\text{escaping out})} = 0.188$, $X_{NO_2(\text{escaping out})} = 0.812$

18. $$K_P = \frac{P_{NO_2}}{P_{N_2O_4}} = \frac{\left(\frac{1.2}{1.6}\right)^2}{\left(\frac{0.4}{1.6}\right)} = 2.2$$

$\therefore K_P$ does not change.

SECTION – C

1. 

2. 

3. $nCH_2 = CH - CH = CH_2 + nCH_2 = \underset{\displaystyle CN}{\underset{|}{CH}} \xrightarrow{\text{copolymerisation}} \left[CH_2 - CH = CH - CH_2 - CH_2 - \underset{\displaystyle CN}{\underset{|}{CH}} \right]_n$

Mathematics

PART – III

SECTION – A

1. If $[x^3 + x^2 + x + 1] = [x^3 + x^2 + 1] + x$, then x is integer
$\Rightarrow \log |[x]| = 2 - |[x]|$ has same solutions as $\log |x| = 2 - |x|$, x is integer
$\Rightarrow$ No integral solution $\Rightarrow 0$

2. Each element has 5 choices, as it may be in any one of A_1, A_2, A_3, A_4 or may not be in any one
$\Rightarrow$ The required number of ways = 5^7

3. $(1 + x)^n = C_0 + C_1x + C_2x^2 \ldots\ldots C_nx^n$

$$\Rightarrow \frac{(1+x)^{n+1}-1}{n+1} = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} \ldots\ldots \frac{C_nx^{n+1}}{n+1}$$

$$\Rightarrow \frac{(1+x)^{n+2}}{(n+1)(n+2)} - \frac{x}{(n+1)} - \frac{1}{(n+1)(n+2)} = \frac{C_0x^2}{2\cdot 1} + \frac{C_1x^3}{3\cdot 2} \ldots\ldots \frac{C_nx^{n+2}}{(n+1)(n+2)}$$

Put x = 2, we get $\frac{3^{n+2}-1}{(n+1)(n+2)} - \frac{2}{(n+1)} = \frac{C_0\cdot 2^2}{2\cdot 1} + \frac{C_1\cdot 2^3}{2\cdot 3} \ldots\ldots \frac{C_n\cdot 2^{n+2}}{(n+1)(n+2)}$

$$\Rightarrow \frac{C_0\cdot 2}{2} + \frac{C_1\cdot 2^2}{2\cdot 3} + \ldots\ldots + \frac{C_n\cdot 2^{n+1}}{(n+1)(n+2)} + \frac{1}{n+1} + \frac{1}{2(n+1)(n+2)} = \frac{3^{n+2}}{2(n+1)(n+2)}$$

$$\Rightarrow S_n = \frac{3^{n+2}}{2(n+1)(n+2)}$$

$$\Rightarrow \frac{S_7}{S_6} = \frac{3^9\times 2\times 7\times 8}{2\times 8\times 9\times 3^8} = \frac{7}{3}$$

4. It is given that $|z| = 1$
$z = \cos\theta + i\sin\theta$ (let), $\theta \in (0, 2\pi)$
Now, $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ gives $2|\cos 2\theta| = 1$

$$\Rightarrow \cos 2\theta = \frac{1}{2}, -\frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$ i.e. 8 values

5. As $\cot\theta - \tan\theta = 2\cot 2\theta$
$\Rightarrow \tan\theta = \cot\theta - 2\cot 2\theta$

$$\Rightarrow \frac{1}{2}\tan\frac{\theta}{2} + \frac{1}{4}\tan\frac{\theta}{4} = \frac{1}{4}\cot\frac{\theta}{4} - \cot\theta$$

6. Latus rectum, $4a = 2\sqrt{2}$

$$\Rightarrow a = \frac{1}{\sqrt{2}}$$

Vertex = $\left(\frac{1}{2}, \frac{1}{2}\right)$
Axis = x – y = 0
As origin lies outside the parabola
$\Rightarrow$ Focus = (1, 1), Directrix = x + y = 0

$\Rightarrow$ Equation of parabola is $\left|\frac{x+y}{\sqrt{2}}\right|^2 = (x-1)^2 + (y-1)^2$

$\Rightarrow x^2 + y^2 - 2xy - 4x - 4y + 4 = 0$

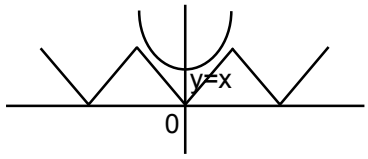
7. Tangent at (0, 0) will be same
$(3 + \sin B)x + (2\cos\alpha)y = 0$ and $2\cos\alpha\, x + 2cy = 0$ are same
$\Rightarrow c = \frac{2\cos^2\alpha}{3+\sin\beta} \Rightarrow c_{max} = 1$ where $\sin\beta = -1$ and $\alpha = 0$

8. Consider $g(x) = xf(x)$
g is continuous on [0, 1] and differentiable on (0, 1)
$\because f(1) = 0 \Rightarrow g(0) = 0 = g(1)$
By Rolles' Theorem $g'(c) = 0$ some for $c \in (0, 1)$
$\Rightarrow cf'(c) + f(c) = 0$

9. At x = 2, RHD = 5 + |1 – x| = 5 + 1 = 6 whereas LHD = 5

11. Let the coordinates of point A are (ct, c/t)
So, the slope of normal at A will be t^2.
And normal will be parallel to BC.
So, t will be $\pm\sqrt{2} \Rightarrow c = \pm 2$.

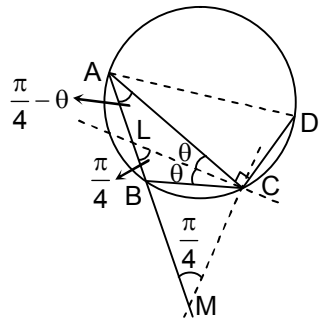
12. 'a' must be negative and $x^2 - a = x$ should have no solution
Now, $D < 0 \Rightarrow 1 + 4a < 0 \Rightarrow a < -\frac{1}{4}$



13. $\because \frac{\Delta ABC}{\Delta PQR} = \frac{b}{a}$
$\Rightarrow \Delta ABC = \frac{b}{a}\cdot\frac{3\sqrt{3}}{4}a^2 = \frac{3\sqrt{3}}{4}ab$

14. Slope of OP and OQ will be $\frac{am_1}{b}$ and $\frac{am_2}{b}$ respectively
If $\angle POQ = \theta$ than $\tan\theta = \left|\frac{\frac{a}{b}(m_1 - m_2)}{1 + \frac{a^2}{b^2}m_1m_2}\right| = \left|\frac{ab(m_1 - m_2)}{b^2 + a^2m_1m_2}\right|$
Now, area of sector OPQ of circle $x^2 + y^2 = a^2 = \frac{\pi a^2}{2\pi}\theta = \frac{1}{2}a^2\theta$
Area of sector OAB = $\frac{b}{a}\left(\frac{1}{2}a^2\cdot\theta\right) = \frac{1}{2}ab\theta$ where $\theta = \tan^{-1}\left|\frac{ab(m_1 - m_2)}{b^2 + a^2m_1m_2}\right|$

15.-16. $\because$ $\angle LCM = 90^\circ$ and $CL = CM$



$\angle CLM = \angle CML = \frac{\pi}{4}$

Let $\angle ACB = 2\theta$

$\Rightarrow \angle LAC = \frac{\pi}{4}-\theta$ and $\angle LCB = \frac{3\pi}{4}-\theta$

By sine Rule in ΔABC

$a^2 + b^2 = 4R^2\left(\sin^2\left(\frac{\pi}{4}-\theta\right)+\sin^2\left(\frac{3\pi}{4}-\theta\right)\right)$

$a^2 + b^2 = 4R^2\left(\sin^2\left(\frac{\pi}{4}-\theta\right)+\cos^2\left(\frac{\pi}{4}-\theta\right)\right) = 4R^2$

Let $\angle ADC = \alpha \Rightarrow \angle ABC = \pi - \alpha \Rightarrow \angle CBM = \alpha$

$\Rightarrow \alpha = \frac{\pi}{4}+\theta$

$\because$ $\angle CAD = \frac{\pi}{2}-\alpha = \frac{\pi}{4}-\theta$

$\therefore \angle BAC = \angle CAD \Rightarrow BC = CD$

17.-18. We want to have $a_n = k$ if $\frac{k(k-1)}{2} < n \le \frac{k(k+1)}{2}$

$\because$ n is an integer, this is equivalent to $\frac{k(k-1)}{2}+\frac{1}{8} < n < \frac{k(k+1)}{2}+\frac{1}{8}$

$\Rightarrow k^2 - k + \frac{1}{4} < 2n < k^2 + k + \frac{1}{4}$

$\Rightarrow k - \frac{1}{2} < \sqrt{2n} < k + \frac{1}{2} \Rightarrow k < \sqrt{2n} + \frac{1}{2} < k + 1$

Hence, $a_n = \left[\sqrt{2n}+\frac{1}{2}\right] \Rightarrow \alpha = 2 = \beta$

17. Now, a = 2, b = 3, c = 5

Let A = number of numbers which are divisible by 2

B = number of numbers which are divisible by 3

C = number of numbers which are divisible by 5

Required number = $A + B + C - A \cap B - B \cap C - C \cap A + A \cap B \cap C$

$= \left[\frac{1000}{2}\right]+\left[\frac{1000}{3}\right]+\left[\frac{1000}{5}\right]-\left[\frac{1000}{6}\right]-\left[\frac{1000}{15}\right]+\left[\frac{1000}{10}\right]+\left[\frac{1000}{30}\right] = 734$

18. a = 2, b = 3, c = 5, d = 7

Hence, the given number is $2^5 \cdot 3^5 \cdot 5^3 \cdot 7^3$

$\because$ 4n + 1 is odd number therefore the factor 2 will not occur in divisor. 3 and 7 are of 4n + 3 form, odd powers of 3 and 7 will be of 4n + 3 form and even powers will be 4n + 1 form

5 is 4n + 1 form and any power of 5 will be of 4n + 1 form

$\therefore$ Number of divisors of 4n + 1 type

= number of terms in the product $(1 + 3^2 + 3^4)(1 + 5 + 5^2 + 5^3)(1 + 7^2)$

+ number of terms in the product $(3 + 3^2 + 3^5)(1 + 5 + 5^2 + 5^3)(7 + 7^3) = 48$

SECTION – C

1. As g(x) is inverse of f(x)
$\Rightarrow$ Domain of g(x) is range of f(x)
$\Rightarrow$ a = 3, b = 11

2. Required number of ways = co-efficient of x^{30} in $(x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + x^8)^5$
= co-efficient of x^{30} in $\left(\frac{x^2(1-x^7)}{1-x}\right)^5$
= co-efficient of x^{20} in $(1 - x^7)^5(1 - x)^{-5}$
$= {}^{24}C_{20} - 5^{17}C_{13} + 10^{10}C_6 = 826$
$\Rightarrow$ A = 8, B = 2, C = 6
A + B – C = 4

3. $\sqrt{\left(\sqrt{x^3+1}-1\right)^2} + \sqrt{\left(\sqrt{x^3+1}-3\right)^2} = \left|\sqrt{x^3+1}-1\right| + \left|\sqrt{x^3+1}-3\right|$
For $x \in [0, 2]$ $1 \le \sqrt{x^3+1} \le 3$
$\Rightarrow$ f(x) = 2
$\therefore \int_0^2 f(x)\,dx = 4$

4. Orthocentre is foot of perpendicular drawn from origin on the plane